

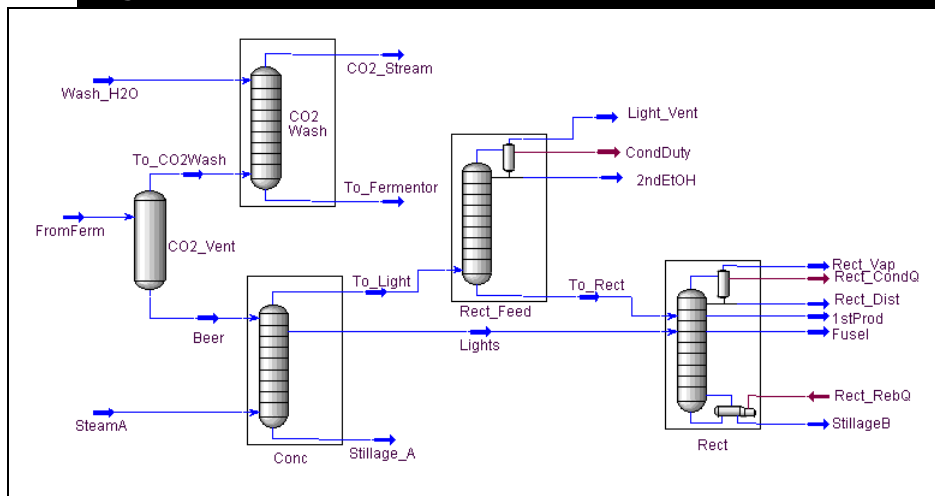
C1 Ethanol Plant

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C1.1 Process Description

Typically, an ethanol fermentation process produces mainly Ethanol plus small quantities of several by-products: methanol, 1-propanol, 2-propanol, 1-butanol, 3-methyl-1-butanol, 2-pentanol, acetic acid, and CO₂.

Figure C1.1

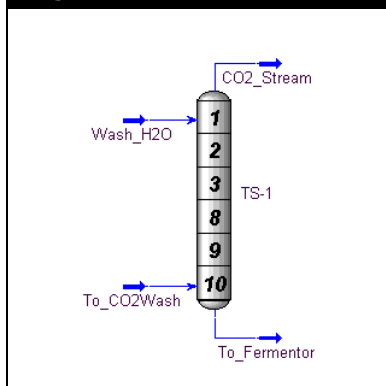


To see this case completely solved, see your HYSYS\Samples\ directory and open the C-2.hsc file.

Ethanol and Water form an azeotropic mixture at 1 atm. Therefore, with simple distillation, the ethanol and water mixture can only be concentrated up to the azeotropic concentration.

The CO_2 produced in the fermentation vessel carries some ethanol. This CO_2 stream is washed with water in a vessel (CO₂ Wash) to recover the Ethanol, which is recycled to the fermentor.

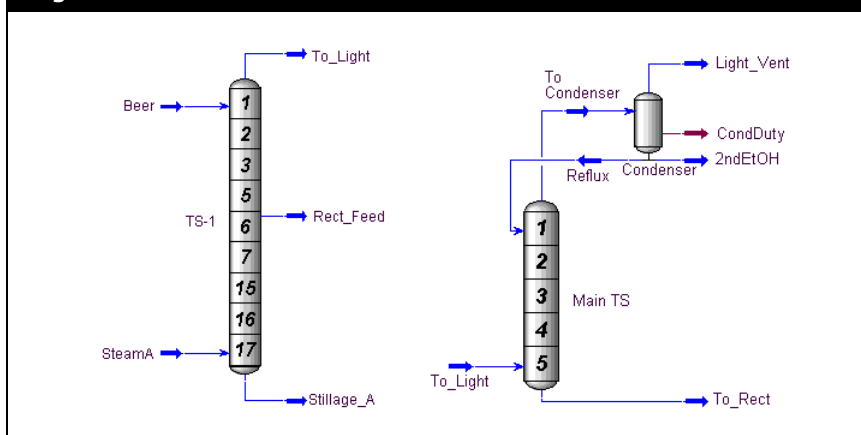
Figure C1.2



The Ethanol rich product stream from the fermentor is sent to a concentration (Conc) tower. An absorber with a side vapour draw can be used to represent this tower.

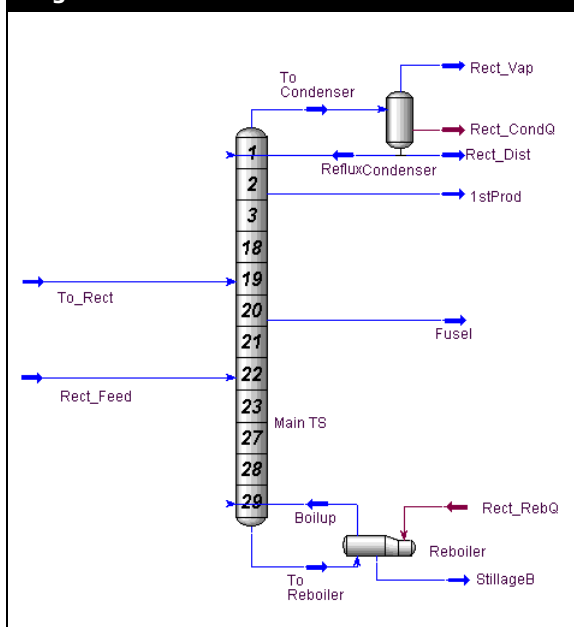
The top vapour is fed to a light purification tower (Lights) where most of the remaining CO_2 and some light alcohols are vented. The bottom product of this light tower is fed to the Rectifier.

Figure C1.3



The side vapour draw from the Concentrator is the main feed for the Rectifier. The Rectifier is operated as a conventional distillation tower. The product of this tower is taken from Stage 2 so to have an azeotropic ethanol product with a lesser methanol contamination. Methanol concentrates towards the top stages, so a small distillate draw is provided at the condenser. Also, a small vent for CO₂ is provided at the condenser.

Figure C1.4



Another factor of interest is the concentration of heavy alcohols in the interior of the Rectifier. These alcohols are normally referred to as Fusel oils, and a small side liquid draw is provided in the Rectifier to recover these components.

Fusel oils are a mixture of propanols, butanols, and pentanols that have a potential value superior to that of ethanol. Accumulation of fusel oils in the Rectification Tower can cause the formation of a second liquid phase and subsequent deterioration of performance for these trays, so small side liquid draws of fusel oils are installed on the rectifier to avoid this problem.

There are two general steps in this process simulation:

1. **Setup.** The NRTL property package and the UNIFAC VLE estimation method will be used for this case. The Components list includes Ethanol, H₂O, CO₂, Methanol, Acetic Acid, 1- Propanol, 2-Propanol, 1-Butanol, 3-M-1-C4ol, 2-Pentanol and Glycerol.
2. **Steady State Simulation.** This case will use a separator, two absorbers, a refluxed absorber and a distillation column.

C1.2 Setup

1. In the Session Preferences property view, set the unit set to **SI**.
2. In the Component List property view, select the following components: **Ethanol, H₂O, CO₂, Methanol, Acetic Acid, 1- Propanol, 2-Propanol, 1-Butanol, 3-M-1-C4ol, 2-Pentanol**, and **Glycerol**.
3. In the Fluid Package property view, select **NRTL** as the property package.
4. On the **Binary Coeffs** tab of the Fluid Package property view, select the **UNIFAC VLE** radio button and click the **Unknowns Only** button to estimate the missing interaction parameters.

C1.3 Steady State Simulation

C1.3.1 Adding Streams

Enter the Simulation environment and add the material streams defined below.

Name	Wash H2O	FromFerm	Steam A
In this cell...	Enter...	Enter...	Enter...
Temperature [C]	25	30	140
Pressure [kPa]	101.3250	101.3250	101.3250
Molar Flow [kgmole/hr]	130	2400	

Name	Wash H2O	FromFerm	Steam A
Mass Flow [kg/hr]			11000
Comp Mole Frac [Ethanol]	0.0000	0.0269	0.0000
Comp Mole Frac [H2O]	1.0000	0.9464	1.0000
Comp Mole Frac [CO2]	0.0000	0.0266	0.0000
Comp Mole Frac [Methanol]	0.0000	2.693e-05	0.0000
Comp Mole Frac [Acetic Acid]	0.0000	3.326e-06	0.0000
Comp Mole Frac [1-Propanol]	0.0000	9.077e-06	0.0000
Comp Mole Frac [2-Propanol]	0.0000	9.096e-06	0.0000
Comp Mole Frac [1-Butanol]	0.0000	6.578e-06	0.0000
Comp Mole Frac [3-M-1-C4ol]	0.0000	2.148e-05	0.0000
Comp Mole Frac [2-Pentanol]	0.0000	5.426e-06	0.0000
Comp Mole Frac [Glycerol]	0.0000	6.64e-06	0.0000

Once you have entered the Mole Fractions for the stream FromFerm, the Mole Fractions will not add up to 1.00. Click the Normalize button and the total Mole Fraction will equal 1.00.

C1.3.2 Installing Equipment

CO2 Vent Separator

The CO2Vent Separator separates the products from the fermentor. The bottom liquid of the separator is sent to the distillation section of the plant (Concentrator Tower), while the overhead vapour goes to the CO2Wash Tower.

Install a Separator and make the connections shown below.

SEPARATOR [CO2 Vent]		
Tab [Page]	In this cell...	Enter...
Design [Connections]	Inlets	FromFerm
	Vapour Outlet	To CO2Wash
	Liquid Outlet	Beer

CO₂ Wash Tower

Water is used to strip any Ethanol entrained in the off gas mixture, thus producing an overhead of essentially pure CO₂. The bottom product from the tower is recycled to the Fermentor, however, the recycle is not a concern in this example.

1. Before installing the column, select **Preferences** from the HYSYS **Tools** menu.
2. On the **Options** page of the **Simulation** tab, ensure that the **Use Input Experts** checkbox is selected, then close the property view.
3. Install the CO₂ Wash Tower as a simple Absorber.

Absorber [CO2WASH]		
Tab [Page]	In this cell...	Enter...
Connections	No. of Stages	10
	Top Stage Inlet	Wash H2O
	Bottom Stage Inlet	To CO2Wash
	Ovhd Vapour	CO2 Stream
	Bottoms Liquid	To fermentor
Pressure Profile	Top Stage	101.325 kPa
	Bottom Stage	101.325 kPa

4. Click the **Run** button in the Column property view to calculate the CO₂ Wash Tower product streams.

Concentrator

1. Install the Concentrator as an Absorber with a side vapour draw.

Absorber [CONC]		
Tab [Page]	In this cell...	Enter...
Connections	No. of Stages	17
	Top Stage Inlet	Beer
	Bottom Stage Inlet	Steam A
	Ovhd Vapour	To Light
	Bottoms Liquid	Stillage A
	Side Draw Vapour	Rect Feed (Stage 6)
Pressure Profile	Condenser	101.325 kPa
	Reboiler	101.325 kPa
Temperature Estimates	Condenser Temperature	90°C
	Reboiler Temperature	110°C

2. Create and define the following specifications to fully specify the column.

Specifications		
Tab [Page]	In this cell...	Enter...
Design [Specs]	Comp Recovery	Active
	Draw	Rect Feed
	SpecValue	0.95
	Component	Ethanol
	Draw Rate 1	Estimate
	Draw	Rect Feed
	Flow Basis	Mass
	Spec Value	5000 kg/h
	Draw Rate 2	Estimate
	Draw	To_Light
	Flow Basis	Molar
	Spec Value	1000 kgmole/h

You might have to deactivate the default Rect Feed Rate specification to converge the column.

3. Click the **Run** button in the Column property view to calculate the Concentrator product streams.

Lights

1. Add the Lights Tower purification tower, modeled as a Refluxed Absorber, and define as indicated below.

Refluxed Absorber [Lights]		
Tab [Page]	In this cell...	Enter...
Connections	No. of Stages	5
	Bottom Inlet Streams	To Light
	Condenser Type	Partial
	Ovhd Vapour	Light Vent
	Ovhd Liquid	2ndEtOH
	Bottoms Liquid	To Rect
	Cond. Energy	CondDuty
Pressure Profile	Condenser Pressure	101.325 kPa
	Reboiler Pressure	101.325 kPa

2. Delete the default **Btms Prod Rate** and **Reflux Rate** specifications from the Column Specification group.
3. Add the following new column specifications (**Design** tab, **Specs** page).

Specifications		
Tab [Page]	In this cell...	Enter...
Design [Specs]	Vap Prod Rate	Active
	Draw	Light_Vent
	Flow Basis	Molar
	Spec Value	1.6 kgmole/hr
	Comp Fraction	Active
	Stage	Condenser
	Flow Basis	Mass Fraction
	Phase	Liquid
	Spec Value	0.88
	Component	Ethanol
	Reflux Ratio	Estimate
	Stage	Condenser
	Flow Basis	Molar
	Spec Value	5.00
	Distillate Rate	Estimate
	Draw	2ndEtOH
	Flow Basis	Molar
	Spec Value	2.10 kgmole/hr

4. If required, click the **Run** button in the Column property view to calculate the Light Tower product streams.

Rectifier

The primary product from a plant such as this would be the azeotropic mixture of ethanol and water. The Rectifier serves to concentrate the water/ethanol mixture to near azeotropic composition. The Rectifier is operated as a conventional distillation tower. It contains a partial condenser as well as a reboiler.

1. Add the Rectifier column, modeled as a distillation tower, and define it using the following information.

Column [RECT]		
Tab [Page]	In this cell...	Enter...
Connections	No. of Stages	29
	Inlet Streams [Stage]	To Rect [19], Rect_Feed [22]
	Condenser Type	Partial
	Ovhd Vapour	Rect Vap
	Ovhd Liquid	Rect Dist
	Bottoms Liquid	Stillage B
	Reboiler Energy	Rect RebQ
	Condenser Energy	Rect CondQ
	Side Draw Liquid [Stage]	1st Prod [2], Fusel [20]
Pressure Profile	Condenser Pressure	101.325 kPa
	Reboiler Pressure	101.325 kPa
Temperature Estimates	Condenser	79°C
	Reboiler	100°C

2. Delete the default **Btms Prod Rate** and **Reflux Rate** specifications before adding the new specifications. Delete all specifications that do not appear in the following table.

3. Define the following specifications (**Design** tab, **Specs** page). Also, set the damping factor to accelerate the convergence.

Specifications		
Tab [Page]	In this cell...	Enter...
Design [Specs]	Reflux Ratio Stage Flow Basis Spec Value	Active Condenser Molar 7100
	Ovhd Vap Rate Draw Flow Basis Spec Value	Active Rect_Vap Molar 0.100 kgmole/hr
	Draw Rate Draw Flow Basis Spec Value	Active Rect_Dist Mass 2.00 kg/hr
	Comp Frac Stage Flow Basis Phase Spec Value Component	Active 2_Main TS Mass Fraction Liquid 0.95 Ethanol
	Fusel Draw Rate Draw Flow Basis Spec Value	Active Fusel Mass 3.00 kg/hr
	1stProd Draw Rate Draw Flow Basis Spec Value	Estimate 1stProd Molar 68.00 kgmole/hr
Parameters [Solver]	Damping Factor Fixed	0.25
	Azeotrope Check	ON

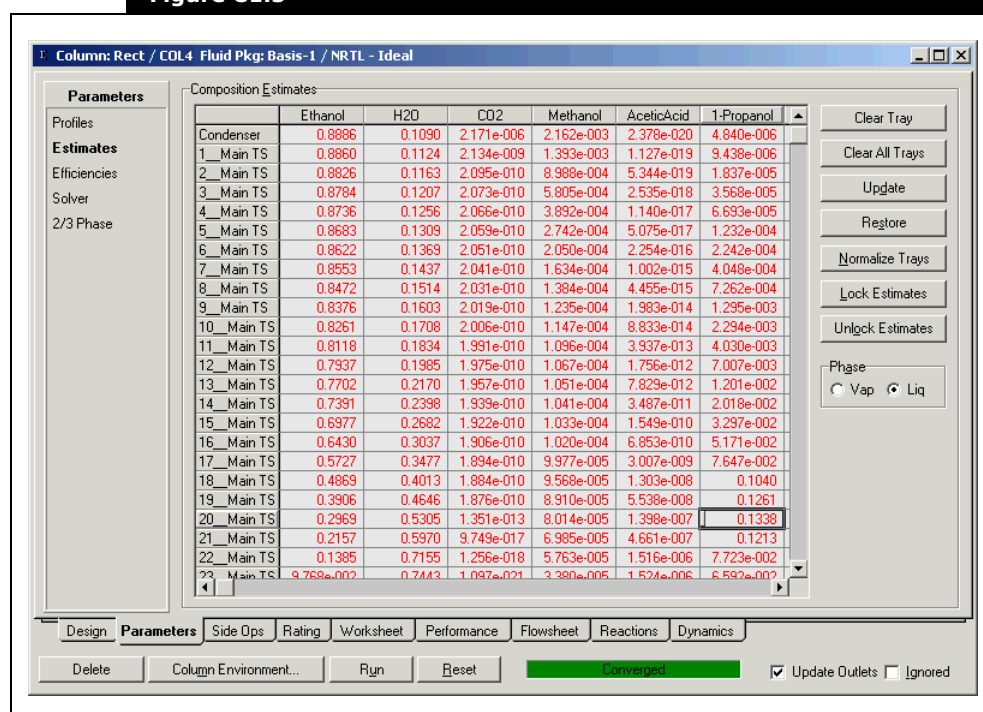
4. Click the **Run** button to solve the column.

C1.3.3 Draw Stream Location

The side liquid draw, Fusel, is added at stage 20. To determine if this is an appropriate stage to recover the heavy alcohols, view the stage-by-stage composition profile.

1. To examine this information, click the **Parameters** tab in the Column property view.
2. Select the **Estimates** page. In this property view you can see the Composition Estimates of each tray.

Figure C1.5



3. To view the 1-Propanol composition on Tray 20, scroll through the group until you can see Tray 20 and the 1-Propanol component.

Stage 20 has a high concentration of 1-Propanol (which has the greatest concentration among the heavy alcohols). Therefore, we have selected the appropriate stage for the Fusel draw.

C1.4 Results

Workbook Case (Main)

- Material Streams Tab

Figure C1.6

Name	Wash H2O	FromFerm	Steam A	To CO2 Wash	Beer	CO2 Stream
Vapour Fraction	0.0000	0.0277	1.0000	1.0000	0.0000	1.0000
Temperature [C]	25.00	30.00	140.0	30.00	30.00	26.06
Pressure [kPa]	101.3	101.3	101.3	101.3	101.3	101.3
Molar Flow [kgmole/h]	130.0	2400	610.6	66.47	2334	64.72
Mass Flow [kg/h]	2342	4.672e+004	1.100e+004	2857	4.386e+004	2792
Liquid Volume Flow [m3/h]	2.347	48.16	11.02	3.454	44.70	3.375
Heat Flow [kJ/h]	-3.704e+007	-6.892e+008	-1.448e+008	-2.557e+007	-6.636e+008	-2.515e+007

Name	To Fermenter	To Light	Stillage A	Rect Feed	Light Vent	2ndEtOH
Vapour Fraction	0.0000	1.0000	0.0000	1.0000	1.0000	0.0000
Temperature [C]	33.34	85.89	100.3	94.82	46.40	46.40
Pressure [kPa]	101.3	101.3	102.3	101.6	101.3	101.3
Molar Flow [kgmole/h]	131.8	10.03	2617	316.6	1.600	2.687
Mass Flow [kg/h]	2407	301.9	4.716e+004	7402	68.89	104.8
Liquid Volume Flow [m3/h]	2.426	0.3511	47.25	8.126	8.379e-002	0.1285
Heat Flow [kJ/h]	-3.745e+007	-2.556e+006	-7.308e+008	-7.506e+007	-5.678e+005	-7.444e+005

Name	To Rect	Rect Vap	Rect Dist	Stillage B	1st Prod	Fusel
Vapour Fraction	0.0000	1.0000	0.0000	0.0000	0.0000	0.0000
Temperature [C]	80.85	78.01	78.01	99.73	78.09	83.39
Pressure [kPa]	101.3	101.3	101.3	101.3	101.3	101.3
Molar Flow [kgmole/h]	5.742	0.1000	4.653e-002	253.1	69.04	8.660e-002
Mass Flow [kg/h]	128.2	4.306	2.000	4566	2955	3.000
Liquid Volume Flow [m3/h]	0.1388	5.360e-003	2.489e-003	4.578	3.675	3.541e-003
Heat Flow [kJ/h]	-1.602e+006	-2.324e+004	-1.261e+004	-7.068e+007	-1.871e+007	-2.435e+004

Material Streams Compositions Energy Streams Unit Ops

FeederBlock_Wash H2O
CO2 Wash

☐ Include Sub-Flowsheets
☐ Show Name Only
Number of Hidden Objects: 0

- Compositions Tab

Figure C1.7

Workbook - Case (Main)

Name	Wash H2O	FromFerm	Steam A	To CO2 Wash	Beer	CO2 Stream
Comp Mole Frac (Ethanol)	0.0000	0.0269	0.0000	0.0170	0.0272	0.0000
Comp Mole Frac (H2O)	1.0000	0.9464	1.0000	0.0409	0.9722	0.0333
Comp Mole Frac (CO2)	0.0000	0.0266	0.0000	0.9421	0.0005	0.9667
Comp Mole Frac (Methanol)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Comp Mole Frac (AceticAcid)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Comp Mole Frac (1-Propanol)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Comp Mole Frac (2-Propanol)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Comp Mole Frac (1-Butanol)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Comp Mole Frac (3-M-1-C4ol)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Comp Mole Frac (2-Pentanol)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Comp Mole Frac (Glycerol)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Name	To Fermentor	To Light	Stillage A	Rect Feed	Light Vent	2ndE10H
Comp Mole Frac (Ethanol)	0.0086	0.3164	0.0000	0.1903	0.1890	0.7451
Comp Mole Frac (H2O)	0.9910	0.5610	1.0000	0.8091	0.0517	0.2524
Comp Mole Frac (CO2)	0.0005	0.1213	0.0000	0.0000	0.7589	0.0007
Comp Mole Frac (Methanol)	0.0000	0.0002	0.0000	0.0002	0.0001	0.0004
Comp Mole Frac (AceticAcid)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Comp Mole Frac (1-Propanol)	0.0000	0.0003	0.0000	0.0001	0.0001	0.0006
Comp Mole Frac (2-Propanol)	0.0000	0.0003	0.0000	0.0001	0.0002	0.0008
Comp Mole Frac (1-Butanol)	0.0000	0.0001	0.0000	0.0000	0.0000	0.0000
Comp Mole Frac (3-M-1-C4ol)	0.0000	0.0001	0.0000	0.0002	0.0000	0.0000
Comp Mole Frac (2-Pentanol)	0.0000	0.0003	0.0000	0.0000	0.0000	0.0001
Comp Mole Frac (Glycerol)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Name	To Rect	Rect Vap	Rect Dist	Stillage B	1st Prod	Fusel
Comp Mole Frac (Ethanol)	0.1513	0.8883	0.8886	0.0002	0.8826	0.2969
Comp Mole Frac (H2O)	0.8473	0.1057	0.1090	0.9995	0.1163	0.5305
Comp Mole Frac (CO2)	0.0000	0.0024	0.0000	0.0000	0.0000	0.0000
Comp Mole Frac (Methanol)	0.0001	0.0034	0.0022	0.0000	0.0009	0.0001
Comp Mole Frac (AceticAcid)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Comp Mole Frac (1-Propanol)	0.0003	0.0000	0.0000	0.0000	0.0000	0.1338
Comp Mole Frac (2-Propanol)	0.0001	0.0002	0.0002	0.0000	0.0003	0.0003
Comp Mole Frac (1-Butanol)	0.0003	0.0000	0.0000	0.0001	0.0000	0.0035
Comp Mole Frac (3-M-1-C4ol)	0.0002	0.0000	0.0000	0.0002	0.0000	0.0000

Material Streams **Compositions** Energy Streams Unit Ops

FeederBlock_Wash H2O
CO2 Wash

☐ Include Sub-Flowsheets
☐ Show Name Only
Number of Hidden Objects: 0

- Energy Streams Tab

Figure C1.8

Workbook - Case (Main)

Name	CondOutly	Rect Reb Q	Rect CondQ
Heat Flow [kJ/h]	3.584e+005	2.764e+007	4.043e+007
Name	** New **		
Heat Flow [kJ/h]			

Material Streams Compositions **Energy Streams** Unit Ops

Lights

☐ Include Sub-Flowsheets
☐ Show Name Only
Number of Hidden Objects: 0